

Recursive Offline Distillation via Epistemic Complexity Approximation

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Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Quite frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate offline data complexity with a single-token answer entropy and use a medium model’s (70B) entropy as an aleatoric proxy. Next, we apply selective distillation only for a small subset of the reasoning chains with the highest corresponding epistemic component. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve (matching baseline score on 15% of the data). Additionally, we present comparative analysis of methods to generate, normalize and augment synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain. We publish our code and data to facilitate further research in this direction.

1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them (GLM-5-Team et al., 2026; Kimi-Team et al., 2026; DeepSeek-AI et al., 2026; Qwen Team, 2026) demonstrate exceptional results on a vast variety of tasks and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026)

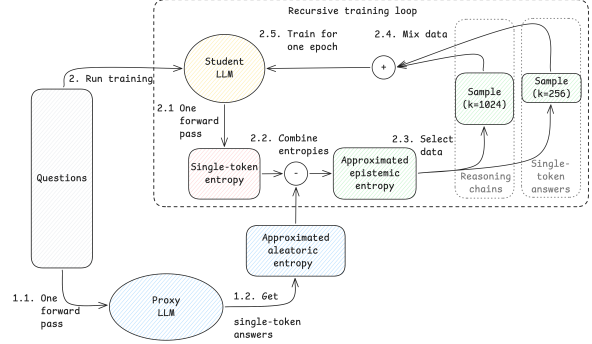


Figure 1: Recursive offline distillation scheme

that can be fine-tuned and deployed in resource-constrained environments.

One of the most prominent approaches for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher, its logits and internal representations (Agarwal et al., 2024; Jiao et al., 2020; Gu et al., 2024), which is not always possible (Zhou et al., 2023; Ryskulov et al., 2026). It forces practitioners to rely on offline distillation, where the teacher model generates reasoning text for a given dataset, and the student model is trained on this data (Li et al., 2023; Hsieh et al., 2023).

Recent work has improved LLM distillation through training objectives (Ko et al., 2024; Xu et al., 2026). Alternatively, data curation (Just et al., 2026; Li et al., 2025; Yang et al., 2026) and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026) have been explored with the scoring criteria remaining a key challenge (Jia et al., 2026).

We propose a fully automated recursive pipeline for offline distillation (Figure 1) based on uncertainty decomposition (Wang et al., 2025): (1) measure the medium model’s entropy once, using it as the aleatoric uncertainty proxy; (2) for each training epoch, estimate the epistemic component, sample the dataset, and run the distillation on the

selected data mix. The method consistently outperforms naive distillation (57.15 ± 1.01 vs 51.61 on average), and it is designed to work in resource-constrained environments.

We validate our approach by fine-tuning three open-source models Qwen2.5 3B (Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024) on a multiple-choice question answering dataset MMLU-Pro (Wang et al., 2024). To confirm how the method works in practice under hardware constraints, we run the experiments on a single A100/H100¹ GPU with 80GB of VRAM.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

2 Related Work

3 Method

Our central hypothesis is that we can approximate epistemic uncertainty of the student model with its entropy and entropy of a larger LLM. Goncharov et al. (2026) show that a single-token entropy (Figure 2) is a reasonable complexity predictor for a given entry in a question-answering dataset. We assume that it approximates the total uncertainty and that we can decompose it into epistemic and aleatoric components (Depeweg et al., 2018; Malinin and Gales, 2021). With scaling laws (Kaplan et al., 2020) in mind, we hypothesize that larger models should have lower total uncertainty. Therefore, in comparison to a small student model (3-4B) its substantially larger sibling (70-72B²) could be treated as a weak aleatoric proxy.

For a question x , assume $H(x) = A(x) + E(x)$, where H_S , H_P - single-token entropy (student and proxy), A - shared aleatoric component, and E_S , E_P - model-specific epistemic component. If proxy’s residual epistemic uncertainty satisfies $E_P(x) \leq \varepsilon(x)$, then

$$\begin{aligned} A(x) &= H_P(x) - E_P(x) \geq H_P(x) - \varepsilon(x) \\ E_S(x) &= H_S(x) - A(x) \leq H_S(x) - H_P(x) + \varepsilon(x) \end{aligned} \quad (1)$$

¹We had to switch cards depending on availability

²This particular size is chosen based on the imposed resource constraints (80GB VRAM) and open model availability

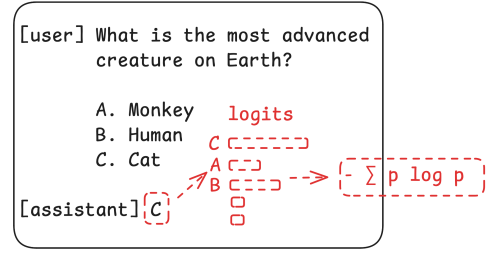


Figure 2: Single-token entropy example

Thus, the entropy difference approximates the student’s epistemic uncertainty with error at most $\varepsilon(x)$.

4 Results

4.1 Experimental Setup

4.2 Main Results

4.3 Synthetic Traces Generation

4.4 Ablation Study

5 Conclusion and discussion

Limitations

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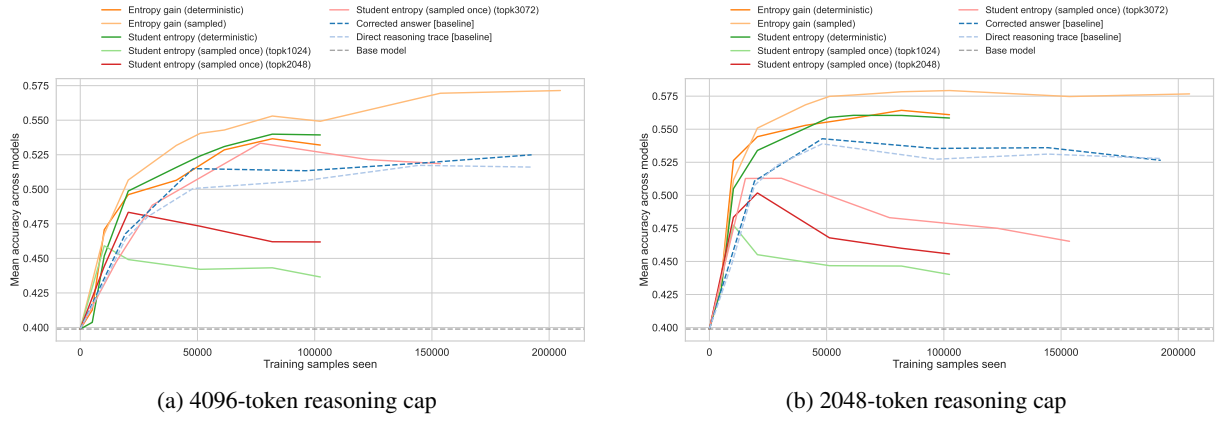


Figure 3: Distillation pipeline results across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro

Method	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	58.81 / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	58.35 / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17 / 50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
	52.58 $\pm$ 1.94 /	56.64 $\pm$ 3.79 /	62.22 $\pm$ 2.42 /	57.15 $\pm$ 1.01 /
Entropy gain (sampled)	52.08 $\pm$ 2.28	56.86 $\pm$ 2.32	64.07 $\pm$ 4.52	57.67 $\pm$ 1.19

Table 1: Distillation pipeline last epoch accuracy (%) across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token / 2,048-token thinking budget. Sampled entropy gain uses three seeds per model and includes $\pm$ 95% confidence-interval. Other methods are deterministic.

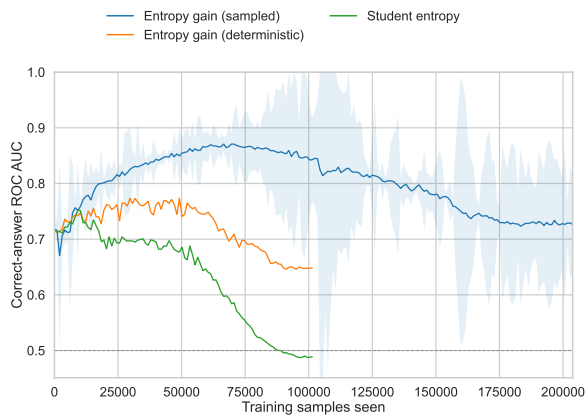


Figure 4: Student calibration

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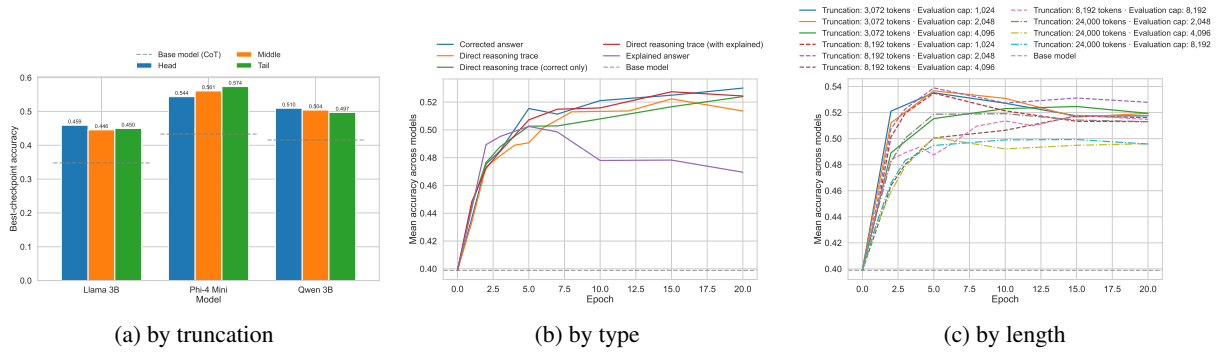


Figure 5: Comparison of synthetic traces

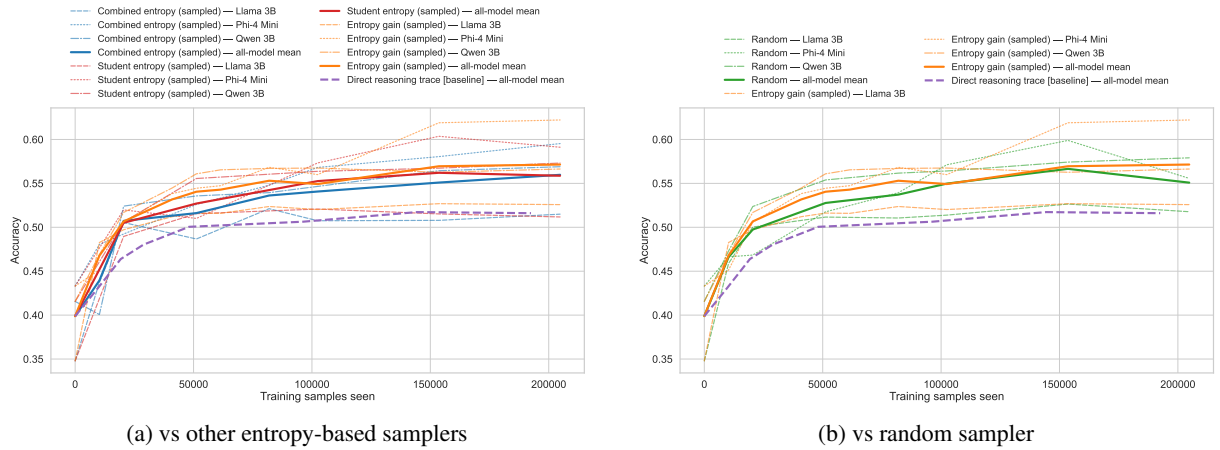


Figure 6: Comparison of entropy gain sampler

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Sampler	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Random	51.77 $\pm$ 2.76	57.91 $\pm$ 1.40	55.57 $\pm$ 2.54	55.08 $\pm$ 0.80
Student entropy	51.18 $\pm$ 3.11	57.32 $\pm$ 1.52	59.12 $\pm$ 5.22	55.87 $\pm$ 1.40
Combined entropy	51.50	56.90	59.53 $\pm$ 2.11	55.98
Entropy gain	52.58 $\pm$ 1.94	56.64 $\pm$ 3.79	62.22 $\pm$ 2.42	57.15 $\pm$ 1.01

Table 2: Distillation pipeline last epoch accuracy for different sampling methods across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token thinking budget. Each method is evaluated with 3 seeds per model and includes $\pm$ 95% confidence-interval.

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A Example Appendix

This is an appendix.